CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

DEPARTMENT OF INFORMATION TECHNOLOGY

A.Y. 2026-27 [EVEN]

LAB MANUAL

DEPSTAR

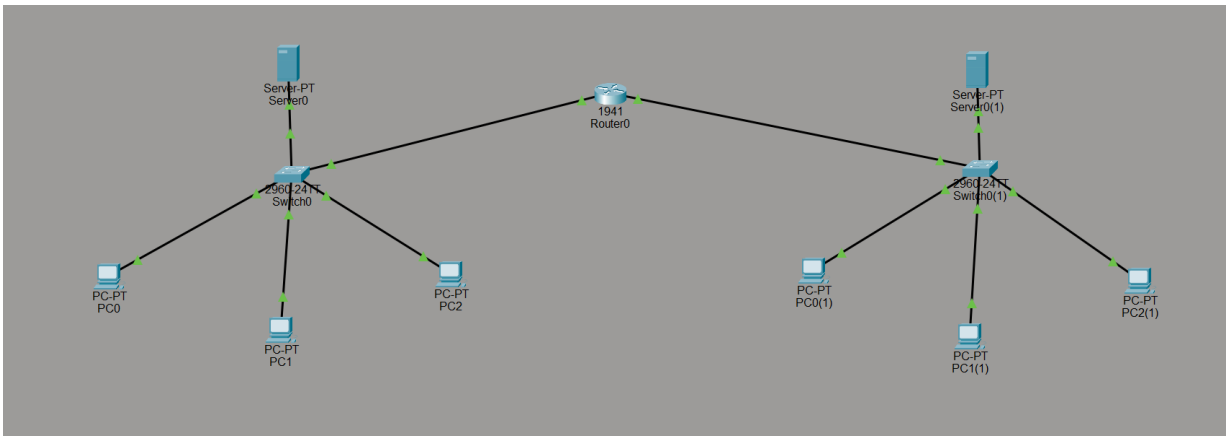
Practical 5

Problem Definition

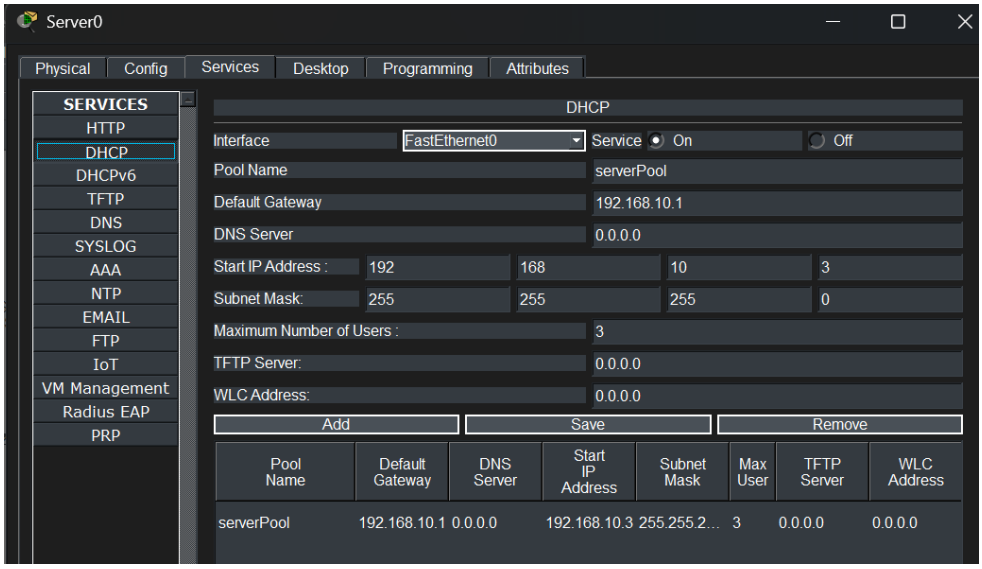
Design and evaluate a multi-branch enterprise network using RIP version 2 dynamic routing and centralized DHCP services. Analyze route advertisement, routing-table updates, DHCP relay operation, DHCP DORA packet exchange, packet forwarding, Layer-2 and Layer-3 address changes, traceroute behavior, and alternate-route selection after introducing a failover link using Cisco Packet Tracer.

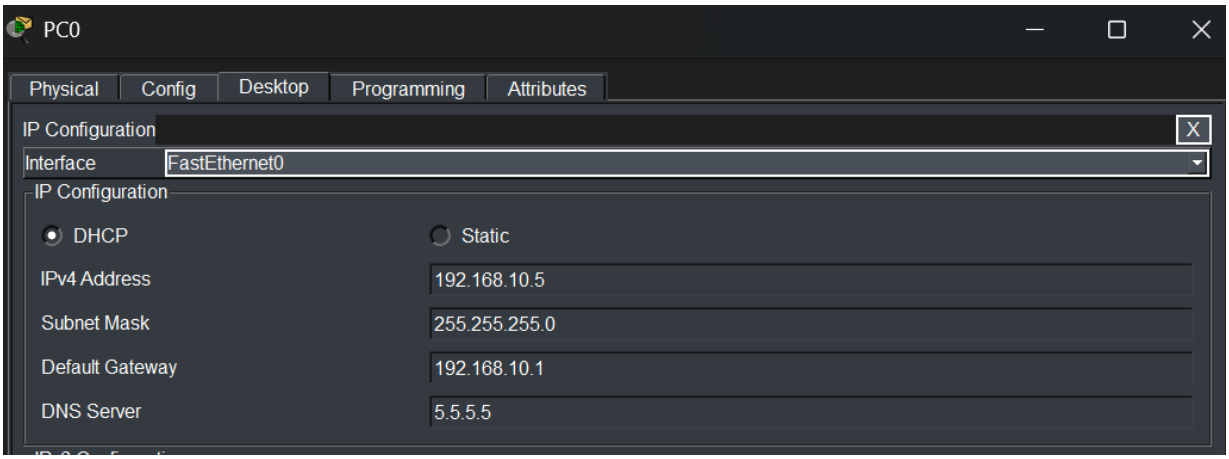
Solution:

Using Different Server



Network-1





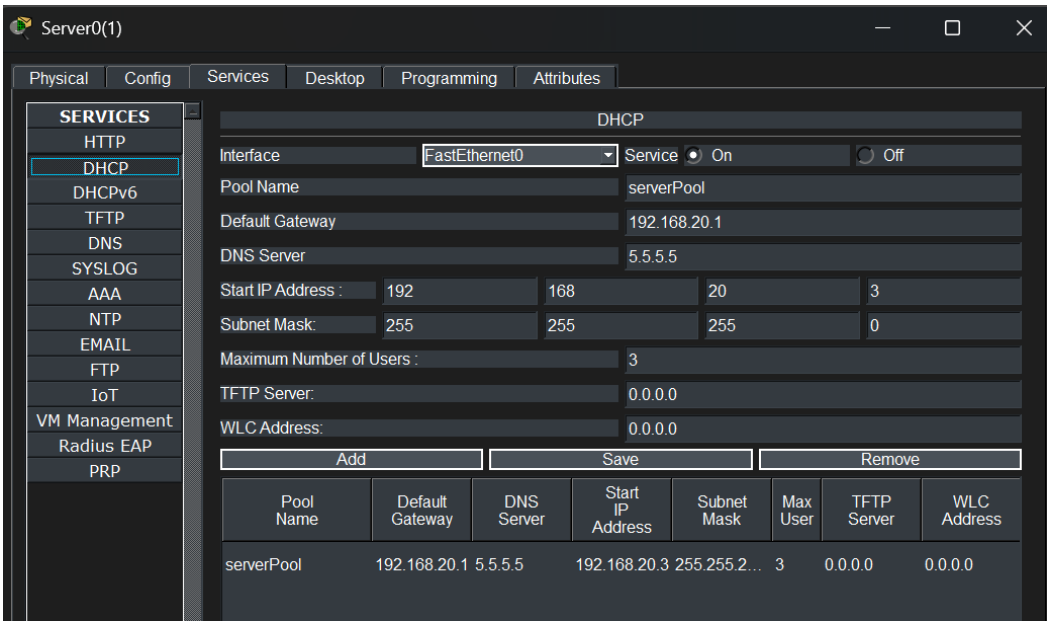

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.4

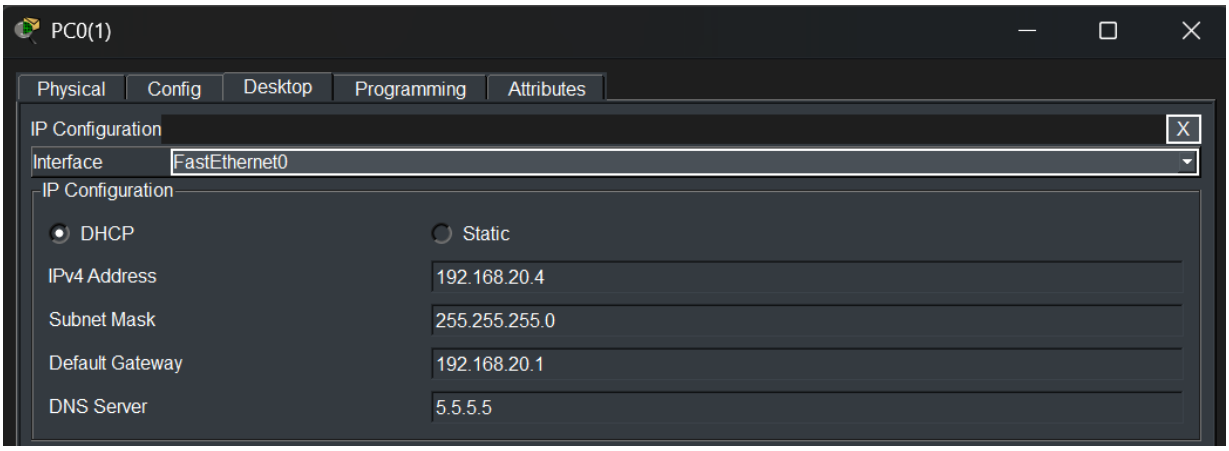
Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Network-2






```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Reply from 192.168.20.5: bytes=32 time<1ms TTL=128
Reply from 192.168.20.5: bytes=32 time=12ms TTL=128
Reply from 192.168.20.5: bytes=32 time=1ms TTL=128
Reply from 192.168.20.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Inter-Network Connection

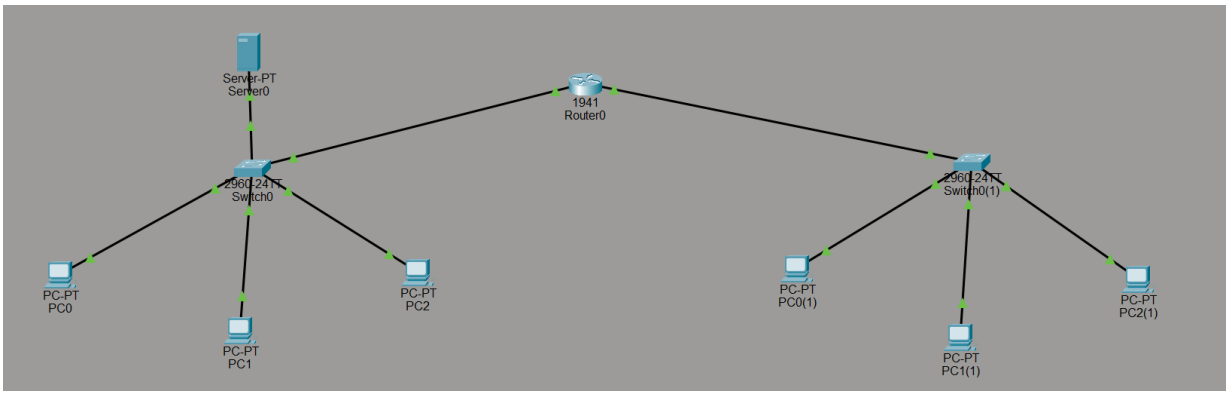
```
C:\>ping 192.168.20.4

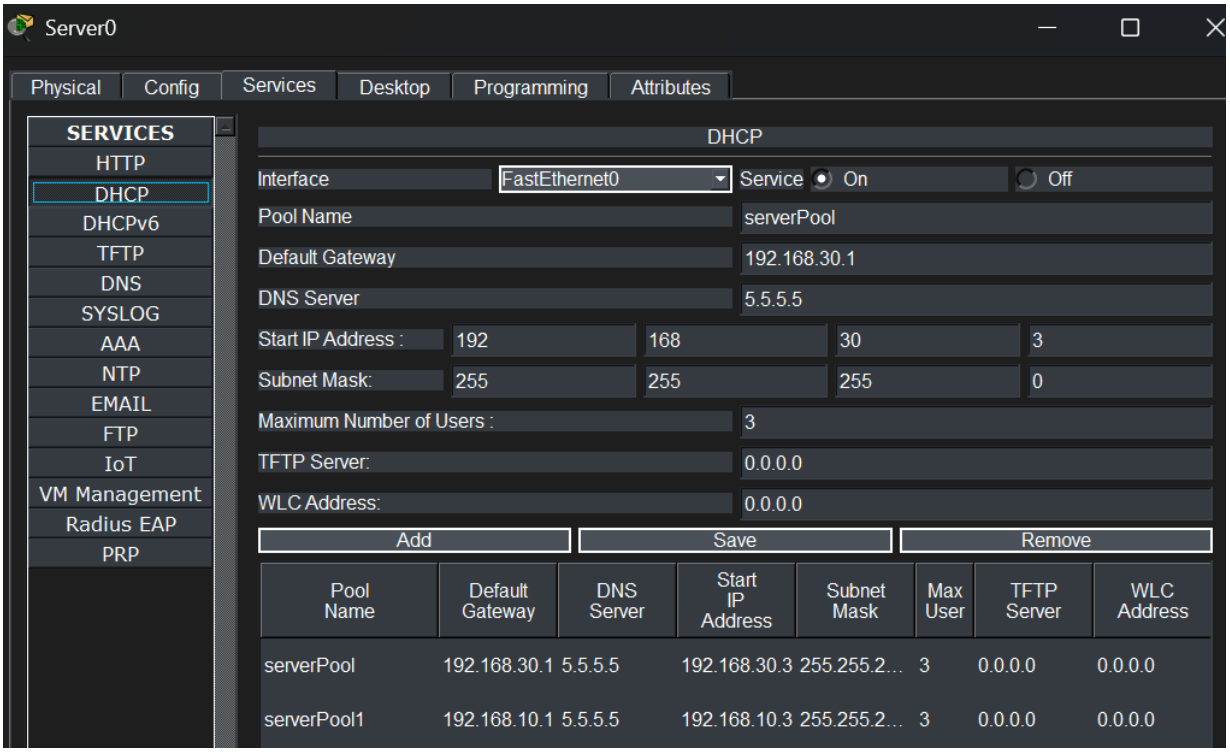
Pinging 192.168.20.4 with 32 bytes of data:

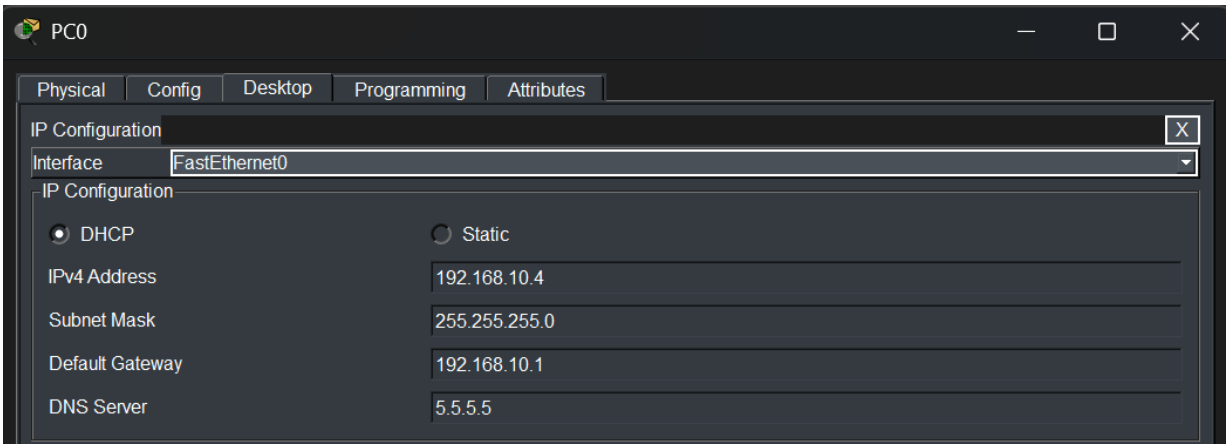
Request timed out.
Reply from 192.168.20.4: bytes=32 time<1ms TTL=127
Reply from 192.168.20.4: bytes=32 time<1ms TTL=127
Reply from 192.168.20.4: bytes=32 time<1ms TTL=127

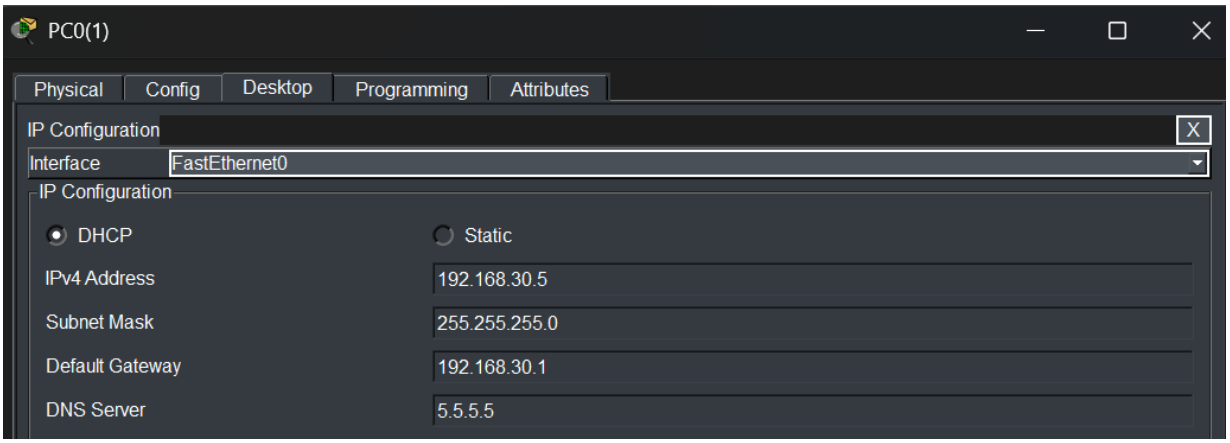
Ping statistics for 192.168.20.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

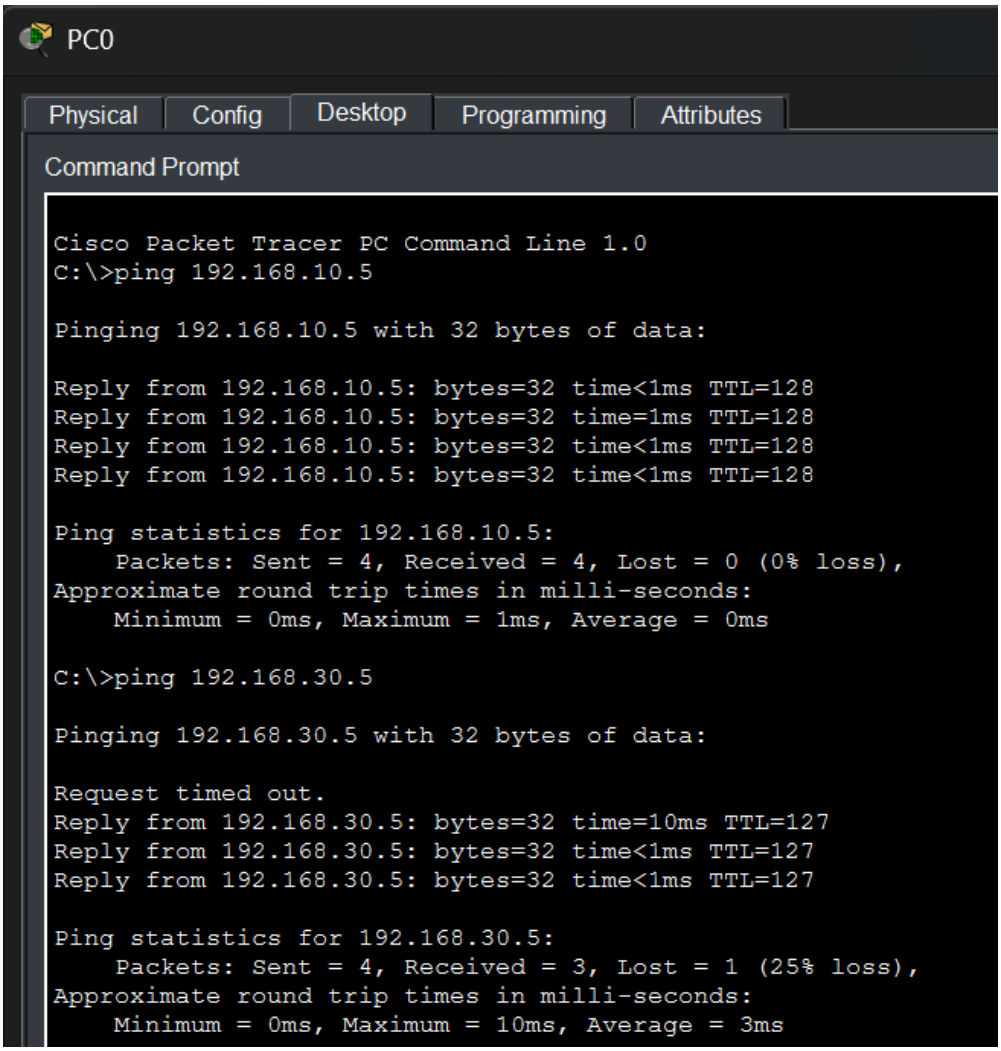
One server


Server-PT
Server0
1941
Router0
2960-24TT
Switch0
2960-24TT
Switch0(1)
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC0(1)
PC-PT
PC1(1)
PC-PT
PC2(1)


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
192.168.30.1
DNS Server
5.5.5.5
Start IP Address :
192
168
30
3
Subnet Mask:
255
255
255
0
Maximum Number of Users :
3
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
192.168.30.1
5.5.5.5
192.168.30.3
255.255.2...
3
0.0.0.0
0.0.0.0
serverPool1
192.168.10.1
5.5.5.5
192.168.10.3
255.255.2...
3
0.0.0.0
0.0.0.0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
5.5.5.5


PC0(1)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.30.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
5.5.5.5


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.5
Pinging 192.168.10.5 with 32 bytes of data:
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.30.5
Pinging 192.168.30.5 with 32 bytes of data:
Request timed out.
Reply from 192.168.30.5: bytes=32 time=10ms TTL=127
Reply from 192.168.30.5: bytes=32 time<1ms TTL=127
Reply from 192.168.30.5: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.30.5:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms

Learning

→ I learned how to configure a multi-branch enterprise network using RIP v2 dynamic routing and centralized DHCP. I understood DHCP DORA, DHCP relay, packet forwarding, Layer-2 and Layer-3 address changes, traceroute, and how routers automatically select an alternate route when the primary link fails